



Traffic Signal Optimization

A Game Theory Approach

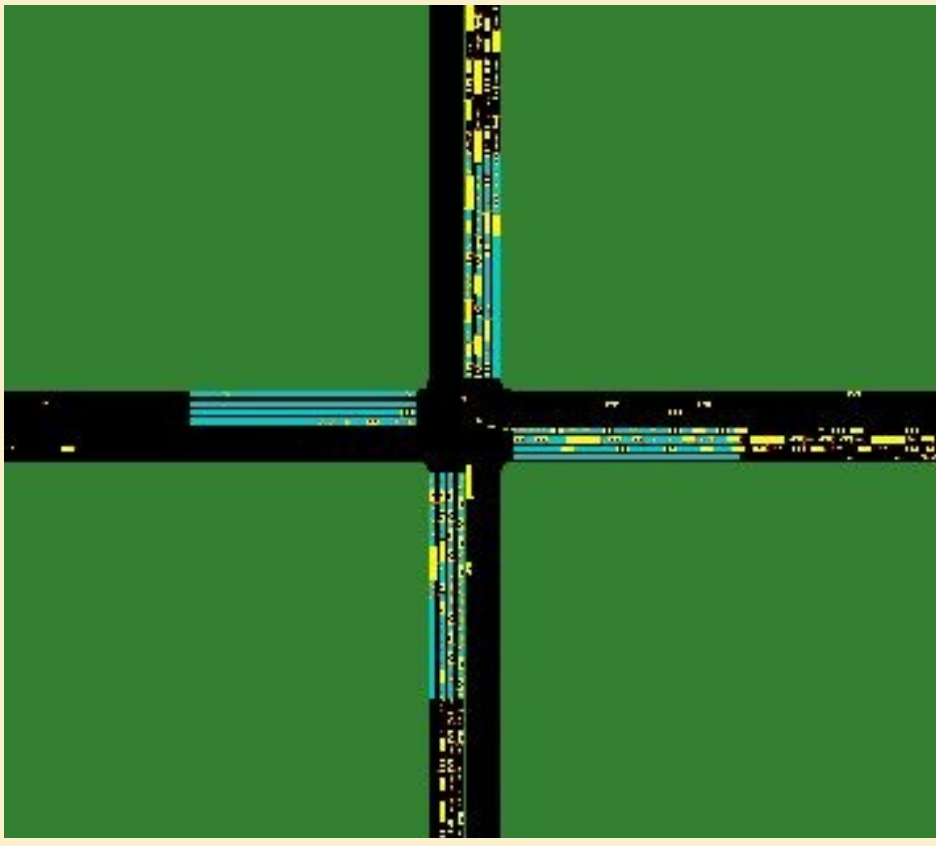
Problem Statement:

“Conventional traffic signals use fixed-time cycles, which are inefficient and cannot adapt to real-time traffic fluctuations, leading to unnecessary congestion and long delays.”

Objective:

- Prove Game Theory solves the traffic problem compared to fixed signals.
- Use Mix and Aggressive Traffic
- Use different times of day(peak and off-peak hours)

Simulation Results:



Before Applying Game Theory
Vehicles Cleared = 38366

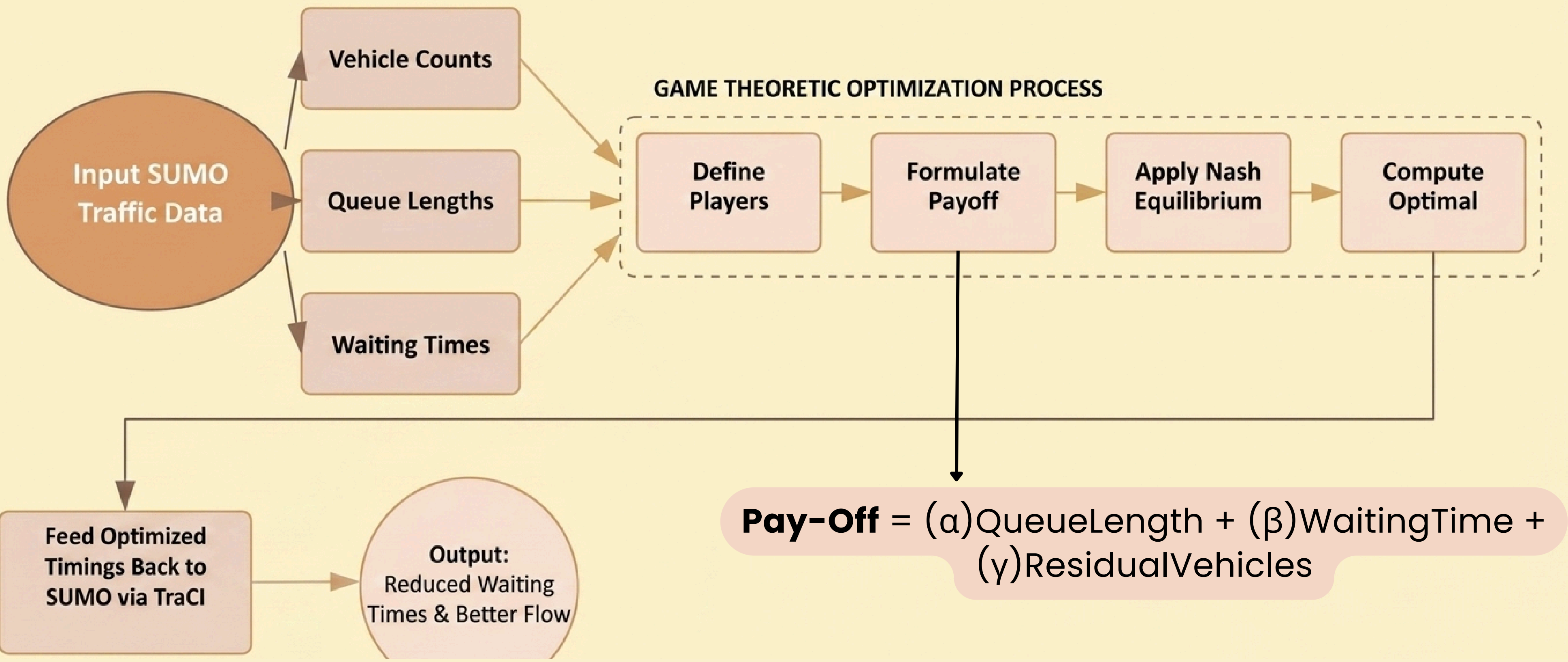


After Applying Game Theory
Vehicles Cleared = 40800

Payoff Matrix:

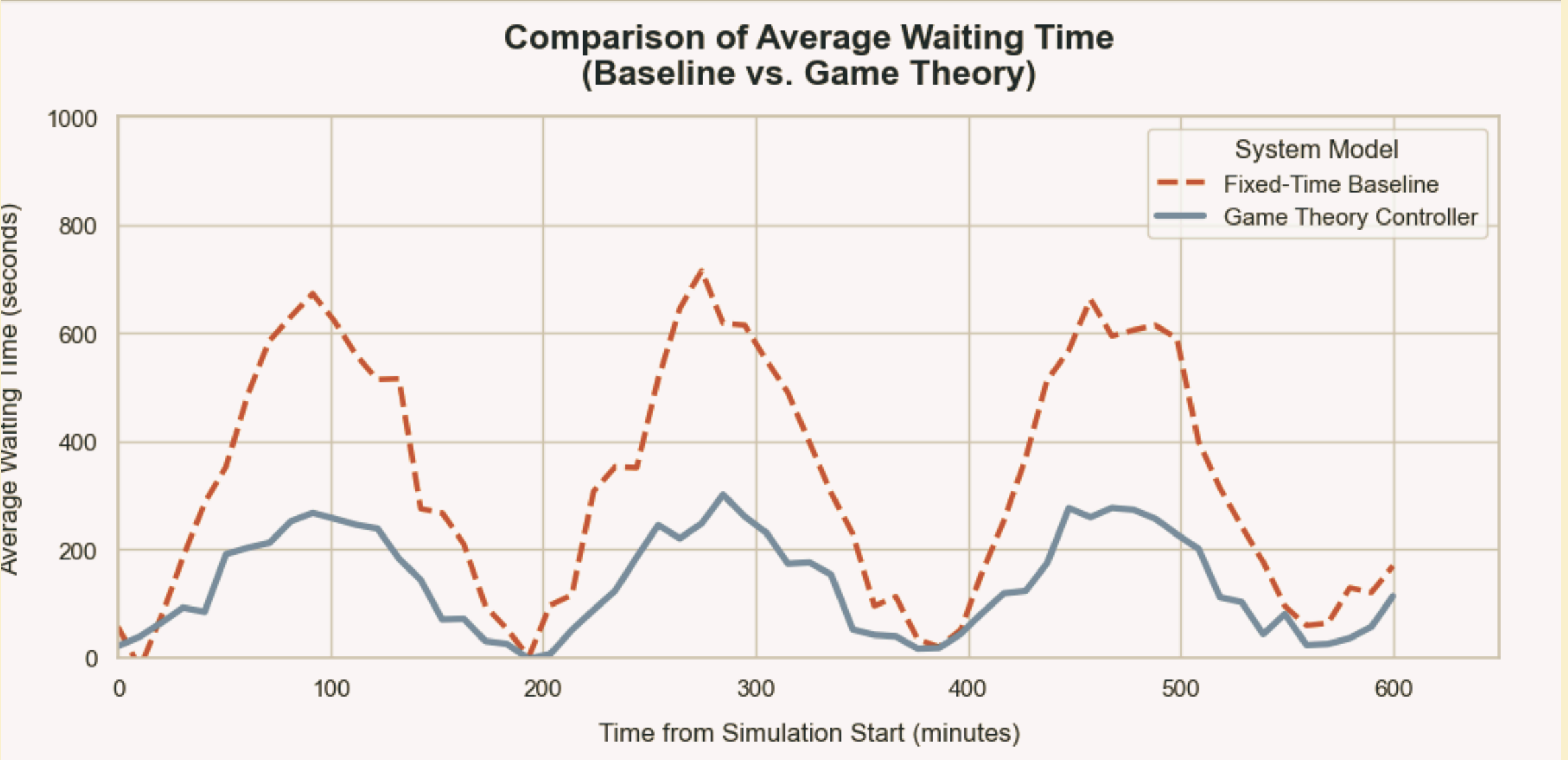
Crash (-10000.0, -10000.0)	NS Goes (0, -297.0)
EW Goes (-748.5, 0)	All Red (-748.5, -297.0)

Traffic Control Logic Flowchart



Results:

- Increased throughput
- Less delays
- Less congestion



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